

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi - 590018



**Internship Report
on**

“MPHASIS METAGEEKS”

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

CSE (IOT & CYBER SECURITY WITH BLOCKCHAIN TECHNOLOGY)

by

NISHA

4MT22IC031

Internship Carried Out at

Mphasis Limited

Under the Guidance of

Internal Guide

**Aneesh Kumar A
Assistant Professor**

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Security with Blockchain Technology)**

External Guide

**Prasanna Ganesh C
Senior QA Engineer
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DEPARTMENT OF CSE (IOT & CYBER SECURITY WITH BLOCKCHAIN TECHNOLOGY)

MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING

(A Unit of Rajalaxmi Education Trust ®, Mangalore)

Autonomous Institute affiliated to VTU, Belagavi, Approved by AICTE, New Delhi

Accredited by NAAC with A+ Grade & ISO 9001:2015 Certified

Institution Badaga Mijar, Moodabidri-574225, Karnataka

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**DEPARTMENT OF CSE (IOT & CYBER SECURITY WITH
BLOCKCHAIN TECHNOLOGY)**

CERTIFICATE

This is to certify that the Internship entitled **“MPHASIS METAGEEKS”** is a bonafide work carried out by **NISHA(4MT22IC031)** in partial fulfillment for the award of degree of **Bachelor of Engineering** in **CSE (IoT & Cyber Security with Blockchain Technology)** of the **Visvesvaraya Technological University, Belagavi during the year 2025-26**. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Internship report has been approved as it satisfies the academic requirements in respect of Internship prescribed for the Bachelor of Engineering degree.

Mr.Aneesh Kumar A
Internal Guide

Dr. Shivananda V Seeri
Head of the Department

Dr. Prashanth C M
Principal

External Viva

Name of the Examiners

Signature with Date

1)

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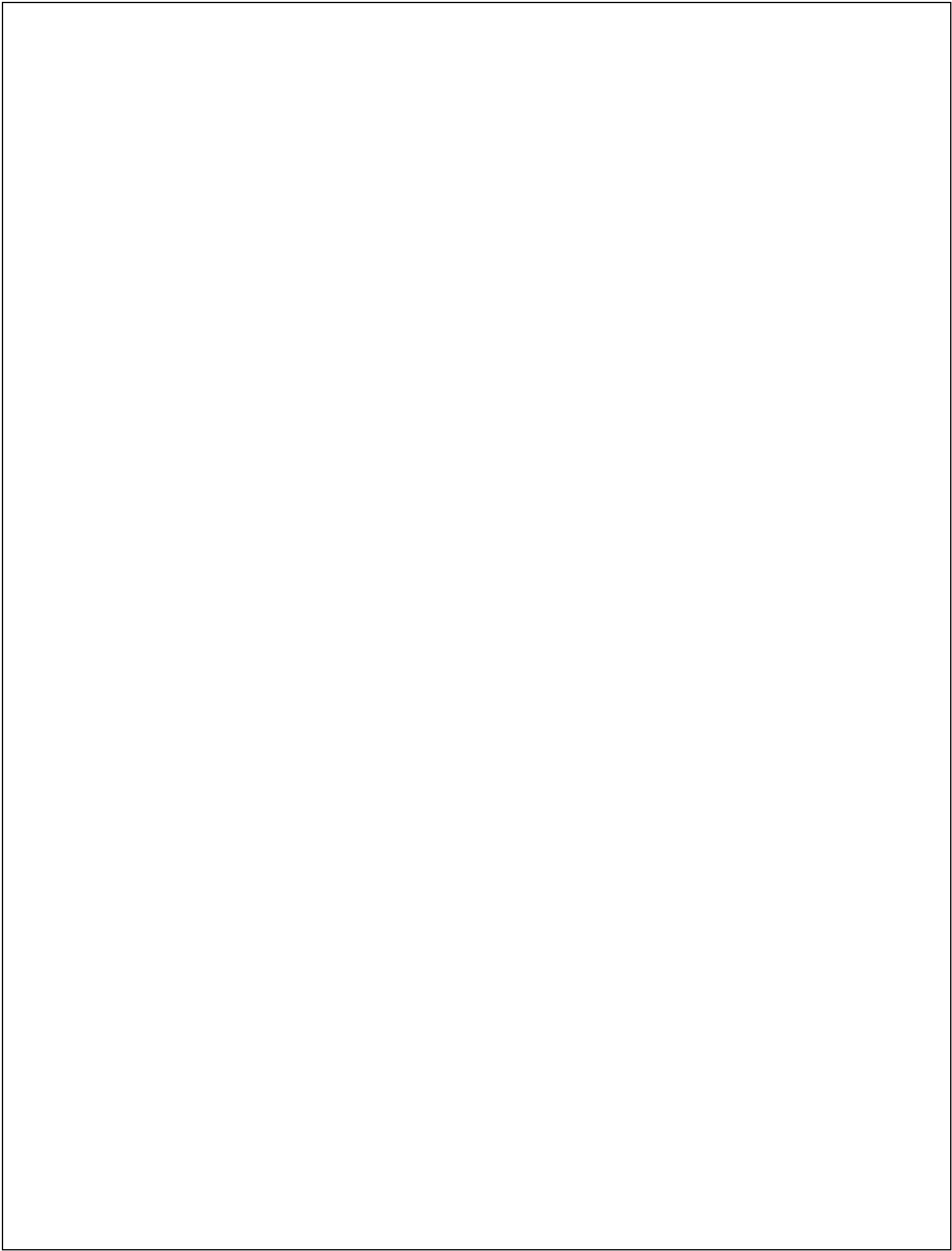
DECLARATION

I, **NISHA(4MT22IC031)**, student of 8th semester BE in **CSE (IoT & Cyber Security with Blockchain Technology) Engineering, Mangalore Institute of Technology and Engineering, Moodabidri**, hereby declare that the Internship entitled “**MPHASIS METAGEEKS**”, submitted to the **Visvesvaraya Technological University, Belagavi** during the academic year **2025-26**, is a report of an original work done by me under the guidance of **Aneesh Kumar A**, Assistant Professor, Dept. of CSE (IoT & Cyber Security with Blockchain Technology) Engineering, Mangalore Institute of Technology and Engineering Moodabidri and **Prasanna Ganesh C**, Senior QA Engineer, Mphasis Limited. This Internship work is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in CSE(IoT & Cyber Security with Blockchain Technology).

Date:

Place: Moodabidri

NISHA



ACKNOWLEDGEMENTS

The success and final outcome of this internship required a lot of guidance and assistance from many people and I am extremely privileged to have got this all along the completion of my internship. All that I have done is only due to such supervision and assistance and I will not forget to thank them.

I would like to thank my Internal Internship Guide **Aneesh Kumar A**, Assistant Professor, Department CSE(IoT & Cyber Security with Blockchain Technology) for his constant encouragement and support extended throughout.

I would like to thank my External Internship Guide **Prasanna Ganesh C**, Senior QA Engineer, Mphasis Limited for her constant help and guidance extended throughout.

I express my sincere thanks to my Department Internship Coordinator, **Mr. Sandeep Seetaram Naik**, for his valuable suggestions and guidance throughout the duration of the internship.

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I express my gratitude and respect to our Principal **Dr. Prashanth C M**, for encouraging me and granting permission to carry out the internship.

I also thank our management who helped me directly or indirectly in the completion of this Internship.

My special thanks to faculty members and others for their constant help and support.

Above all, I extend my sincere gratitude to my parents and friends for their constant encouragement with moral support.

NISHA

ABSTRACT

The internship at Mphasis provided practical exposure to real-world software testing and quality assurance practices followed in the IT industry. The internship focused on understanding and implementing various testing methodologies including manual testing, automation testing, API testing, regression testing, and defect management. The primary objective of the internship was to bridge the gap between academic learning and industrial practices by providing hands-on experience with modern testing tools and frameworks used in professional software development environments. During the internship, automation testing was performed using Selenium WebDriver with Python and the PyTest framework. Automation frameworks were developed using the Page Object Model (POM) design pattern to improve code maintainability and reusability. API testing was carried out using Postman for validating REST APIs, HTTP request methods, response codes, and response data. Additional activities included smoke testing, regression testing, debugging automation scripts, generating test execution reports using pytest-html, and defect tracking using Jira. Test cases and execution results were managed using TestRail within an Agile and Scrum-based workflow. The internship significantly improved technical knowledge, analytical thinking, problem-solving ability, communication skills, and understanding of professional software testing processes. Overall, the experience provided valuable industry exposure and strengthened the foundation required for future career opportunities in software testing and quality assurance domains.

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WEEKLY OVERVIEW OF INTERNSHIP ACTIVITIES

WEEK 1	DATE	DAY	DESCRIPTION OF THE WORK DONE
	10/02/2026	Tuesday	Worked on JavaScriptExecutor and automation handling techniques.
	11/02/2026	Wednesday	Learned Chrome DevTools Protocol and network interceptor basics.
	12/02/2026	Thursday	Practiced handling cookies and ChromeOptions in Selenium automation.
	13/02/2026	Friday	Studied REST API concepts and HTTP methods such as GET and POST.
	14/02/2026	Saturday	Learned REST Assured basics and implemented demo API programs.
	15/02/2026	Sunday	Learned Selenium IDE, WebDriver, and Selenium Grid basics.
WEEK 2	DATE	DAY	DESCRIPTION OF THE WORK DONE
	16/02/2026	Monday	Studied BDD Cucumber concepts and Gherkin syntax implementation.
	17/02/2026	Tuesday	Worked on step definitions and Cucumber feature file execution.
	18/02/2026	Wednesday	Learned hooks, scenario outlines, and background implementation in Cucumber.
	19/02/2026	Thursday	Practiced Maven commands and dependency management concepts.
	20/02/2026	Friday	Learned Maven build lifecycle and Surefire plugin execution.
	21/02/2026	Saturday	Studied advanced Cucumber reporting and Jenkins basics.
	22/02/2026	Sunday	Learned Selenium IDE, WebDriver, and Selenium Grid basics.
WEEK 3	DATE	DAY	DESCRIPTION OF THE WORK DONE
	23/02/2026	Monday	Learned Postman basics and performed API testing exercises.
	24/02/2026	Tuesday	Worked on Postman assertions and response validations.
	25/02/2026	Wednesday	Learned JWT authentication and mock server concepts in API testing.
	26/02/2026	Thursday	Continued hands-on practice with API testing using Postman.
	27/02/2026	Friday	Started learning UFT One and AI testing concepts.
	28/02/2026	Saturday	Practiced record and playback features in UFT One.
	01/03/2026	Sunday	Practiced Selenium WebDriver setup and browser automation basics.

WEEK 4	DATE	DAY	DESCRIPTION OF THE WORK DONE
	02/03/2026	Monday	Learned Object Repository and Smart Identification concepts in UFT.
	03/03/2026	Tuesday	Worked on checkpoints and debugging techniques in UFT.
	04/03/2026	Wednesday	Practiced VBScript basics including variables and conditions.
	05/03/2026	Thursday	Learned loops and functions implementation in VBScript.
	06/03/2026	Friday	Worked on data-driven testing using Excel and DataTable in UFT.
	07/03/2026	Saturday	Studied framework implementation concepts in UFT automation.
	08/03/2026	Sunday	Worked on browser automation and web element identification techniques.
WEEK 5	DATE	DAY	DESCRIPTION OF THE WORK DONE
	09/03/2026	Monday	Learned API testing and AI testing features in UFT One.
	10/03/2026	Tuesday	Practiced framework design using GenAI concepts in UFT.
	11/03/2026	Wednesday	Started learning performance testing using LoadRunner.
	12/03/2026	Thursday	Learned LoadRunner architecture and protocol concepts.
	13/03/2026	Friday	Practiced recording options and script customization in LoadRunner.
	14/03/2026	Saturday	Learned parameterization and correlation concepts in LoadRunner.
	15/03/2026	Sunday	Learned different Selenium locators such as XPath, ID, and CSS selectors.
WEEK 6	DATE	DAY	DESCRIPTION OF THE WORK DONE
	16/03/2026	Monday	Worked on advanced correlation and debugging techniques.
	17/03/2026	Tuesday	Studied transactions, think time, and response validation concepts.
	18/03/2026	Wednesday	Learned Controller scenarios and LoadRunner report analysis.
	19/03/2026	Thursday	Started learning Confluence and Jira fundamentals.
	20/03/2026	Friday	Practiced creating projects, workflows, and permissions in Jira.
	21/03/2026	Saturday	Worked on configuring boards and managing issues in Jira Cloud.
	22/03/2026	Sunday	Practiced writing basic automation scripts using Selenium with Python.

WEEK 7	DATE	DAY	DESCRIPTION OF THE WORK DONE
	23/03/2026	Monday	Learned sprint planning, epics, and workflow management in Jira.
	24/03/2026	Tuesday	Created Jira and TestRail accounts and explored project management features.
	25/03/2026	Wednesday	Started automation project for invalid login functionality.
	26/03/2026	Thursday	Implemented Page Object Model structure for automation framework.
	27/03/2026	Friday	Worked on Excel-driven data handling for test execution.
	28/03/2026	Saturday	Debugged automation scripts and validated login scenarios.
	29/03/2026	Sunday	Studied TestNG concepts and automation testing workflow.
WEEK 8	DATE	DAY	DESCRIPTION OF THE WORK DONE
	30/03/2026	Monday	Completed invalid login automation task and updated Jira.
	31/03/2026	Tuesday	Started automation task for open savings account module.
	01/04/2026	Wednesday	Created page classes and implemented test scripts for account opening flow.
	02/04/2026	Thursday	Continued automation development and locator implementation.
	03/04/2026	Friday	Performed debugging and execution validation for automation scripts.
	04/04/2026	Saturday	Completed open account automation module and updated Jira tasks.
	05/04/2026	Sunday	Learned Selenium IDE, WebDriver, and Selenium Grid basics.
WEEK 9	DATE	DAY	DESCRIPTION OF THE WORK DONE
	06/04/2026	Monday	Reviewed automation framework structure and reusable components.
	07/04/2026	Tuesday	Worked on API testing validations and response handling.
	08/04/2026	Wednesday	Generated automation execution reports and analyzed test results.
	09/04/2026	Thursday	Practiced regression testing and smoke testing scenarios.
	10/04/2026	Friday	Improved framework organization and reusable utility implementation.
	11/04/2026	Saturday	Worked on defect tracking and issue management using Jira.
	12/04/2026	Sunday	Worked on JavaScriptExecutor and advanced Selenium handling techniques.

WEEK 10	DATE	DAY	DESCRIPTION OF THE WORK DONE
	13/04/2026	Monday	Performed end-to-end testing for banking application modules.
	14/04/2026	Tuesday	Validated automation execution and optimized test scripts.
	15/04/2026	Wednesday	Worked on Agile sprint tasks and updated project progress.
	16/04/2026	Thursday	Improved debugging and synchronization handling in Selenium scripts.
	17/04/2026	Friday	Practiced API validations and authentication handling using Postman.
	18/04/2026	Saturday	Reviewed completed modules and prepared project documentation.
	19/04/2026	Sunday	Practiced writing basic automation scripts using Selenium with Python.
WEEK 11	DATE	DAY	DESCRIPTION OF THE WORK DONE
	20/04/2026	Monday	Worked on automation maintenance and script optimization activities.
	21/04/2026	Tuesday	Updated test cases and execution details in TestRail.
	22/04/2026	Wednesday	Performed final validation and execution of automation test cases.
	23/04/2026	Thursday	Worked on report preparation and internship documentation activities.
	24/04/2026	Friday	Reviewed project deliverables and updated pending Jira tasks.
	25/04/2026	Saturday	Improved automation coverage and validated execution results.
	26/04/2026	Sunday	Learned basics of software testing methodologies and SDLC/STLC concepts.
WEEK 12	DATE	DAY	DESCRIPTION OF THE WORK DONE
	27/04/2026	Monday	Performed framework testing and verified reusable components.
	28/04/2026	Tuesday	Conducted regression testing and defect verification activities.
	29/04/2026	Wednesday	Worked on final project review and documentation updates.
	30/04/2026	Thursday	Verified automation scripts and generated final reports.
	01/05/2026	Friday	Reviewed internship learnings and completed documentation work.
	02/05/2026	Saturday	Worked on final report preparation and technical content review.
	03/04/2026	Sunday	Worked on Selenium locators and web element identification techniques.
WEEK 13	DATE	DAY	DESCRIPTION OF THE WORK DONE
	04/05/2026	Monday	Updated internship diary and project summary documentation.
	05/05/2026	Tuesday	Reviewed automation framework and execution reports.

06/05/2026	Wednesday	Finalized project details and completed report corrections.
07/05/2026	Thursday	Verified all internship tasks and prepared final submission documents.
08/05/2026	Friday	Completed internship documentation and final report submission work.

CHAPTER – 1

ABOUT THE ORGANIZATION

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ABOUT THE ORGANIZATION

1.1 Introduction to the Organization

Mphasis Limited is a leading global Information Technology (IT) solutions and services company that specializes in cloud computing, digital transformation, application services, infrastructure services, and business process outsourcing solutions. The company provides innovative technology-driven services to clients across various industries such as banking, financial services, insurance, healthcare, logistics, retail, and telecommunications. Mphasis was founded in the year 1998 and is headquartered in Bengaluru, Karnataka, India. Over the years, the company has established itself as one of the prominent IT service providers by delivering customer-centric digital solutions and modern software engineering services. The organization focuses on enabling enterprises to transform their business operations through advanced technologies such as Artificial Intelligence (AI), Automation, Cloud Computing, Cybersecurity, Data Analytics, and DevOps. The company follows a “Front2Back Transformation” approach, which aims to improve customer experience by integrating front-end digital technologies with back-end operational systems. Mphasis works with global clients and helps them improve efficiency, reduce operational costs, and accelerate digital innovation. The organization strongly emphasizes innovation, quality, teamwork, and continuous learning. Employees and interns are provided with opportunities to work on real-time projects, modern technologies, and industry-standard practices. During the internship, exposure was gained to software testing methodologies, automation frameworks, API testing tools, and Agile development practices followed within the organization. The internship was carried out in the Quality Assurance (QA) and Software Testing domain, where various testing activities such as manual testing, automation testing, API testing, and defect tracking were performed using industry-standard tools and frameworks.

1.2 Vision, Mission, and Core Values

Vision

The vision of Mphasis is to become a globally recognized leader in delivering next-generation technology solutions and digital transformation services that help enterprises innovate and grow efficiently in a rapidly evolving digital world. The organization aims to provide customer-focused digital solutions by combining technology expertise, innovation, and operational excellence. Mphasis continuously works towards enabling businesses to improve productivity, customer experience, and scalability through modern technologies.

Mission

The mission of Mphasis is to deliver high-quality technology services and business solutions that create long-term value for customers through innovation, reliability, and operational excellence. The company focuses on providing customer-centric digital solutions, enhancing customer satisfaction through quality services, and driving business transformation using modern technologies such as cloud computing, automation, artificial intelligence, and data analytics. Mphasis also emphasizes continuous learning, professional development, teamwork, integrity, ethics, and collaboration while adopting industry best practices, Agile methodologies, and modern automation frameworks to ensure efficient and effective service delivery.

Core Values

Mphasis follows strong organizational values that guide employees and teams in delivering high-quality solutions and maintaining professionalism in the workplace. The organization prioritizes customer requirements by focusing on customer centricity and delivering solutions that improve business value and customer satisfaction. Mphasis encourages innovation and continuous improvement through the adoption of emerging technologies such as Artificial Intelligence, Automation, Cloud Computing, and Data Analytics. The company maintains high standards of integrity, ethics, transparency, and accountability in all business operations and employee activities. It also promotes collaboration and teamwork by encouraging employees to share knowledge and work collectively toward achieving organizational goals. Additionally, Mphasis supports continuous learning and skill development by providing training opportunities that help employees and interns stay updated with modern technologies and industry trends. The organization strongly emphasizes excellence and quality by following best engineering practices, quality assurance processes, and efficient project management methodologies to ensure reliable and effective service delivery.

1.3 Organizational Structure and Key Functions

Mphasis follows a well-defined organizational structure to ensure effective management, smooth communication, and efficient project execution. The organization consists of various departments including software development, quality assurance, business analysis, project management, infrastructure management, and support services. During the internship, work was carried out in the Software Testing and Quality Assurance (QA) Department, where the primary responsibility is to ensure that software applications meet the required quality, reliability, and performance standards before deployment. The QA department performs activities such as manual testing, automation testing, API testing, regression testing, smoke testing, defect tracking, and test case management using industry-standard tools such as Selenium WebDriver, PyTest, Postman, Jira, and TestRail. The department works closely with developers, project managers, and business analysts in an Agile and Scrum environment to identify defects, validate functionalities, and ensure the delivery of high-quality software product.

